

Fairness in Machine Learning Models for Healthcare and Education: A Model-Agnostic Pipeline for Systematic Evaluation and Mitigation

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Abstract—Machine learning models increasingly support high stakes decisions in healthcare, education, and finance, and the EU AI Act classifies many such applications as high risk and requires fairness assessment and safeguards. Existing fairness studies mostly evaluate a single prediction task, treat bias detection and mitigation in isolation, and rarely validate reusable methods across datasets. This paper presents a model-agnostic fairness pipeline that covers detection, mitigation, and reassessment, and characterizes each dataset through four measurable properties: class imbalance, sensitive-attribute cardinality, subgroup base-rate gap, and subgroup sample size. The pipeline is evaluated on six datasets spanning healthcare, education, finance, and income prediction, whose positive prevalence ranges from 0.12 to 0.70. Across the datasets, it executes a uniform matrix of three classifiers, four mitigation methods with one control, and multiple sensitive attributes, and augments standard metrics with two diagnostic criteria that require no additional experiments. In the examined datasets, the lowest-prevalence data show both the largest group disparities and the highest mitigation cost, and constrained in-processing mitigation can collapse toward all-negative prediction when positives are scarce: one configuration scored near zero on every aggregate fairness metric yet produced only a single, incorrect positive prediction across 1,091 test instances. Baseline selection-rate gaps widely exceed the base-rate difference inherent in the data, in the strongest case an output gap of 0.588 against a true gap of 0.133; constraint-based and threshold-based mitigation remove this excess, whereas removal of the sensitive column does not. On multi-category attributes, mitigation conclusions reverse across base classifiers. The pipeline and the two criteria together provide a reusable audit procedure for fairness evaluation in high-risk applications.

Index Terms—algorithmic fairness, bias mitigation, machine learning pipeline, healthcare, education, group disparity

I. INTRODUCTION

With the growth of data availability and computational capacity, machine learning models are deployed in high-stakes decision scenarios at increasing scale [1]. The EU AI Act classifies systems used for medical prediction and student performance evaluation as high-risk applications and explicitly requires assessment and safeguards for their transparency and fairness [2]. In healthcare, model outputs in tasks

such as depression risk prediction, readmission assessment, and disease progression analysis directly influence diagnostic decisions and the allocation of medical resources [3,4,5,6]. Evidence from these tasks shows that such models can amplify existing social inequalities and behave inconsistently across demographic groups [3,6].

Another application where machine learning is becoming increasingly prevalent is education through learning analytics systems that predict student outcomes, identify students at risk of failure or dropping out, and support timely interventions. Similar to healthcare applications, these predictions may influence access to educational support and future academic opportunities, making fairness an important consideration. The Open University Learning Analytics Dataset (OULAD) contains demographic, behavioural, and academic performance information, providing a suitable benchmark for evaluating disparities in predictive performance across different student subgroups [7]. Although educational data mining has traditionally focused on improving predictive accuracy, recent research has highlighted that machine learning models can also exhibit demographic biases, motivating the need for fairness-aware evaluation and mitigation in educational settings [8,9,10].

Both domains shape long-term individual development and social mobility, which makes systematic fairness evaluation unavoidable in these settings.

Existing fairness research has made substantial progress while retaining three limitations. Fairness evaluations are mostly confined to a single prediction task, which makes the stability of a method across application scenarios difficult to judge [4]. Bias detection and mitigation are usually studied in isolation, and work that covers the full detection, mitigation, and reassessment workflow remains limited [3,6]. Some methods are designed to be model-agnostic, yet their reusability across datasets lacks systematic empirical comparison [5]. These limitations point to one gap: a reusable fairness evaluation and mitigation pipeline, validated systematically on multiple datasets.

This paper addresses that gap with a complete, reusable, model-agnostic fairness evaluation and mitigation pipeline whose core evaluation and mitigation modules contain no dataset-specific logic. We apply the pipeline to three datasets with distinct characteristics: NHANES (healthcare, positive prevalence 0.13), Adult Income (0.24), and German Credit (credit scoring, 0.70). We extend the validation to three more datasets: OULAD (education, student performance prediction) and two finance datasets, Bank Marketing and Credit Card Clients. Each dataset is characterized through four measurable properties: the class imbalance of the target label, the cardinality of each sensitive attribute, the base-rate gap between subgroups, and subgroup sample size. The experiments are organized along three research questions. RQ1: how does class imbalance of the target label affect the magnitude of group predictive disparities and the difficulty of their mitigation? RQ2: how does sensitive-attribute cardinality affect the disparity reduction achieved by mitigation, its performance cost, and its stability across base classifiers? RQ3: when subgroups differ in true base rates, do models amplify the gap beyond the base-rate difference, and can mitigation correct the amplified share without performance loss? On top of standard metrics, we formalize two lightweight diagnostic criteria: a False-Fairness Criterion that separates genuine fairness from the disguise of a degenerate model, and a Model-Manufactured Disparity Metric that decomposes an observed gap into a data-inherent share and a model-added share.

The two criteria capture behavior that standard metrics miss: one configuration with near-perfect readings on all three aggregate fairness metrics in fact produced a single positive prediction, itself incorrect, on a 1,091-instance test set, and baseline models turned a true base-rate gap of 0.133 into a selection-rate gap of 0.588.

The contributions of this paper are as follows. (1) A model-agnostic fairness pipeline that covers detection, mitigation, and reassessment, where a new dataset requires only a lightweight adapter. (2) A systematic empirical study on three datasets with distinct characteristics, under a uniform matrix of three classifiers, five treatments, and multiple sensitive attributes, with evidence-based answers to the three research questions. (3) Two diagnostic criteria with zero additional experimental cost, together with a complete account of their trigger conditions and blind spots. (4) A set of findings with direct practical implications, among them the degeneracy risk of constrained mitigation under low prevalence, the classifier dependence of mitigation conclusions on multi-category attributes, and the inability of sensitive-column removal to eliminate model-added disparity.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the pipeline, the dataset characterization, the models and mitigation methods, and the two diagnostic criteria. Section IV reports the experimental results along the three research questions and discusses the underlying mechanisms. Section V concludes with limitations and future work.

II. RELATED WORK

A. Fairness in Machine Learning: Metrics, Mitigation, and Tools

Research on machine learning fairness has developed a mature body of concepts and methods. The survey by Mehrabi et al. [1] maps fairness definitions, sources of bias, and mitigation strategies, and separates bias into a data component, rooted in imbalanced or historically biased training data, and an algorithmic component that further amplifies the former under training and optimization. Real-world cases such as the COMPAS recidivism tool [11] and an automated hiring system [12] demonstrate that algorithmic decisions can carry systematic racial and gender bias. Group fairness is quantified mainly through difference-based metrics: Demographic Parity, also known as statistical parity, requires equal positive prediction rates across groups [13], while equality of opportunity and equalized odds by Hardt et al. [14] define fairness through the alignment of subgroup true positive rates and false positive rates. Mitigation methods divide by intervention stage into pre-processing, in-processing, and post-processing, which adjust the data before training, constrain the learner within training, and modify predictions after training, respectively [1]. Toolkits such as AIF360 [15] and Fairlearn [16] integrate these metrics and algorithms into unified platforms and provide the infrastructure for fairness experiments across models and datasets.

B. Fairness in Healthcare Applications

Healthcare fairness research has evolved from bias identification toward comparative evaluation of mitigation. Yuan et al. [6] study Alzheimer disease progression prediction on ADNI longitudinal data and find that models achieve better gender fairness where the data are relatively balanced on gender, while disparities across racial and ethnic groups persist; the emphasis stays on identification rather than mitigation. Work on multimodal mental health data by Cheong et al. [3] compares pre-processing, in-processing, and post-processing methods, attributes bias to both data distribution and algorithmic design, and reports that methods at different stages diverge strongly in fairness gain and performance impact, with some methods improving minority outcomes at a cost to majority groups. The FAIR-CARE framework [17] evaluates mitigation approaches across multiple medical prediction tasks under one protocol and documents the diversity of fairness-performance trade-offs. Depression prediction across four national cohorts [4] shows that the same model and mitigation method behave differently across populations, and that bias reduction without accuracy loss is achievable under specific conditions. Li et al. [18] import a debugging perspective from software engineering, correct bias features that correlate with sensitive attributes at the pre-processing stage, and evaluate practical feasibility through a favorable decision rate. Raza [5] builds a reusable pipeline with integrated fairness analysis for readmission prediction, but the design depends on particular models and mitigation techniques and lacks validation on other datasets. Across this line of work, multi-dataset and

multi-population settings receive growing attention, yet model-agnostic solutions that cover the full detection, mitigation, and reassessment workflow with cross-dataset validation remain scarce.

C. Fairness in Educational Applications

Research in educational data mining has primarily focused on developing predictive models for student performance, retention, and learning behaviour, with predictive accuracy serving as the main evaluation objective [8]. Early learning analytics studies demonstrated the potential of machine learning to support educational decision-making, but comparatively little attention was given to fairness and the possibility of unequal outcomes across demographic groups. As these models are increasingly used to identify at-risk students and guide educational interventions, ensuring fair treatment across student populations has become an important consideration [9].

The Open University Learning Analytics Dataset (OULAD) is one of the most widely used benchmarks in learning analytics, containing demographic, behavioural, assessment, and academic performance information that makes it well suited for evaluating fairness across student subgroups [7]. Studies have shown that fairness conclusions in student performance prediction depend strongly on the selected fairness definition and metric, meaning that the same model may satisfy one fairness criterion while violating another [10].

Recent work has investigated sources of demographic bias in educational prediction systems and explored mitigation approaches such as oversampling and dataset rebalancing to improve fairness, although these methods often involve trade-offs with predictive performance [19]. Nevertheless, existing research has largely focused on individual prediction tasks or isolated mitigation techniques. Similar to healthcare applications, a reusable workflow that systematically integrates fairness detection, mitigation, and reassessment across multiple educational and non-educational datasets remains lacking.

D. Positioning

Table I compares this paper with representative studies along six dimensions: multi-dataset evaluation, mitigation coverage, fairness-performance trade-off analysis, model-agnostic design, cross-dataset validation, and pipeline integration. Prior work satisfies at most five of the six. This paper covers detection, mitigation, and reassessment in one model-agnostic pipeline, validates it with a uniform experimental matrix on multiple datasets with distinct characteristics, and extends the reading capability of standard metrics through two lightweight diagnostic criteria, which satisfies all six dimensions.

III. METHODOLOGY

A. Pipeline Design

We construct a model-agnostic fairness evaluation and mitigation pipeline that covers the full path from data ingestion to diagnostic reading: preprocessing and sensitive-attribute extraction, model training, fairness mitigation, joint evaluation

TABLE I
COMPARISON WITH REPRESENTATIVE FAIRNESS STUDIES

Study	Multi-dataset	Mitig.	Trade-off	Model-agn.	Cross-ds valid.	Pipeline
Yuan et al. [6]	×	×	×	✓	×	×
Cheong et al. [3]	✓	✓	×	✓	×	×
Criscuolo et al. [17]	✓	✓	✓	✓	×	×
Dang et al. [4]	✓	✓	✓	✓	×	×
Li et al. [18]	✓	✓	✓	✓	✓	×
Raza [5]	×	✓	×	×	×	✓
Baker & Hawn [9]	×	×	✓	✓	×	×
Le Quy et al. [10]	×	×	✓	✓	×	×
Cock et al. [19]	×	✓	✓	×	×	×
Ours	✓	✓	✓	✓	✓	✓

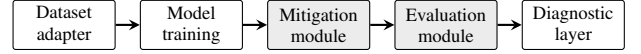


Fig. 1. Pipeline architecture. Shaded modules are dataset-agnostic and frozen across all experiments; each new dataset requires only the adapter.

of performance and fairness, and post hoc diagnostics. Fig. 1 shows the architecture. The core consists of two dataset-agnostic modules: an evaluation module that computes group-level and aggregate metrics on top of Fairlearn MetricFrame, and a mitigation module that wraps one method from each stage of the machine learning lifecycle plus one control. The treatment set includes an untouched Baseline arm, so evaluation of the Baseline serves as detection and evaluation of the mitigated arms serves as reassessment. Neither module contains dataset-specific logic; a new dataset connects through a lightweight adapter responsible for loading, preprocessing, and sensitive-attribute extraction. All datasets share one evaluation protocol, one set of model configurations, and one set of mitigation methods, which keeps results comparable across datasets. The diagnostic layer runs as pure post-processing on the evaluation output, modifies neither training nor evaluation, and adds no experiments.

B. Datasets and Characteristic Profiling

The analysis is not organized by application domain. Each dataset is instead characterized through four measurable properties: class imbalance of the target label, expressed as positive prevalence; the cardinality of each sensitive attribute; the base-rate gap, defined as the difference in true positive-class prevalence between subgroups of an attribute; and subgroup sample size. These properties form the coordinate system that describes and organizes the experiments. Table II summarizes the three core datasets, which span prevalence from 0.13 to 0.70 and cardinality from binary to five categories.

NHANES (healthcare, depression prediction) draws on the NHANES 2011–2023 cycle [20] with an adult subsample of 5,455 participants, a binary depression label constructed from the PHQ-9 questionnaire, and positive prevalence of 0.13, the low-prevalence anchor. Its three sensitive attributes are gender (2 groups), income (3 groups from the family income to poverty ratio, cut at 1.3 and 1.85), and race (5 groups), with base-rate gaps of 0.058, 0.133, and 0.068.

Adult Income [21] predicts whether annual income exceeds 50K on 48,842 records with prevalence of 0.24, the mid-range point. Its sensitive attributes are sex (2 groups, base-rate gap 0.193) and race (5 groups, base-rate gap 0.142). Two race subgroups, Amer-Indian-Eskimo with 96 test instances and Other with 82, provide the conditions to examine how subgroup size affects metric reliability.

German Credit (credit scoring) [22] contains 1,000 records with good credit as the positive class and prevalence of 0.70, the high-prevalence anchor, where positives form the majority and the imbalance direction reverses relative to the other two datasets. Its sensitive attributes are sex (2 groups, base-rate gap 0.048) and age group (2 groups, base-rate gap 0.149), where the young subgroup holds only 28 test instances.

OULAD (education, student performance prediction) [7] contains 32,593 student records with a binary outcome indicating successful course completion and a positive prevalence of 0.47, representing a relatively balanced prediction task. Its sensitive attributes are gender (2 groups, base-rate gap 0.021), region (13 groups, base-rate gap 0.189), and age band (2 groups, base-rate gap 0.144). The combination of demographic diversity and behavioural learning data makes OULAD well suited for evaluating fairness across both binary and multi-category sensitive attributes.

Bank Marketing [23,24] (finance, term-deposit subscription prediction) contains 45,211 client records with a positive prevalence of 0.12, providing a second low-prevalence classification task from a different application domain. Its sensitive attributes are age group (2 groups, base-rate gap 0.008), marital status (2 groups, base-rate gap 0.031), and education (2 groups, base-rate gap 0.054). The relatively small base-rate gaps allow evaluation of whether fairness disparities emerge primarily from the learning algorithm rather than substantial underlying differences between demographic groups.

Credit Card Clients [25,26](finance, default prediction) contains 30,000 records with a positive prevalence of 0.22, representing a moderately imbalanced classification task. Its sensitive attribute is gender (2 groups), with a base-rate gap of 0.034. This dataset complements the other finance datasets by examining fairness in credit default prediction under comparatively small inherent differences between demographic groups.

C. Models and Mitigation Methods

Each dataset trains three classifiers: Logistic Regression, Random Forest, and XGBoost, which cover a linear model, a bagging ensemble, and a boosting ensemble. Class imbalance is handled uniformly at the model level: Logistic Regression and Random Forest use balanced class weights, and XGBoost sets the positive-class weight to the negative-to-positive ratio. Hyperparameters for Random Forest and XGBoost come from randomized search under 5-fold cross-validation (Random Forest: 500 trees, maximum depth 15, minimum leaf size 4; XGBoost: 800 rounds, learning rate 0.01, maximum depth 3), Logistic Regression operates on standardized features, and every stochastic step runs under a fixed random seed.

TABLE II
DATASET CHARACTERISTICS

Dataset	Domain	n	Prev.	Attribute (card.)	Base-rate gap
NHANES	Health	5,455	0.13	gender (2) income (3) race (5)	0.058 0.133 0.068
Adult	Income	48,842	0.24	sex (2) race (5)	0.193 0.142
German	Credit	1,000	0.70	sex (2)	0.048
OULAD	Education	32,593	0.47	age group (2) gender (2) region (13)	0.149 0.021 0.189
Bank Marketing	Finance	45,211	0.12	age band (2) age group (2) marital status (2) education (2)	0.144 0.008 0.031 0.054
Credit Card Clients	Finance	30,000	0.22	gender (2)	0.034

Four treatments cover the three lifecycle stages plus one control, each applied to an arbitrary base classifier:

- 1) **Reweighting** (pre-processing, AIF360) reweights training samples by the joint distribution of the sensitive attribute and the label, with the majority subgroup as the privileged group [27];
- 2) **ExponentiatedGradient** (in-processing, Fairlearn) is a reductions method under an equalized odds constraint, with constraint slack 0.01 and at most 50 iterations [28];
- 3) **ThresholdOptimizer** (post-processing, Fairlearn) learns a separate decision threshold per subgroup from predicted probabilities under an equalized odds constraint with balanced accuracy as the objective, and reserves 20% of the training set as an internal validation split to avoid leakage [14];
- 4) **Suppression** (control) retrains after removal of the sensitive columns and tests the naive assumption that fairness follows from blindness to the sensitive attribute [29].

The experimental matrix per dataset is three classifiers, Baseline plus the four treatments, and all sensitive attributes of that dataset.

D. Fairness Metrics and Evaluation Protocol

Evaluation proceeds on two levels. At the group level, each subgroup of each sensitive attribute reports selection rate, true positive rate, false positive rate, subgroup size, and true positive-class prevalence. At the aggregate level, three difference metrics are reported, each taken as the maximum minus the minimum across subgroups: Demographic Parity difference over selection rates, Equal Opportunity difference over true positive rates, and Equalized Odds difference as the larger of the true positive rate and false positive rate differences.

The protocol is identical for all datasets: a stratified 80/20 train-test split, 5-fold stratified cross-validation on the training set for performance stability, a final fit on the full training set, and all fairness and performance results reported on the

held-out test set. Performance metrics are accuracy, precision, recall, F1, and ROC-AUC. Direct cross-dataset comparison is restricted to the gender attribute, the only sensitive attribute with a comparable structure across datasets; all other attributes are analyzed within their dataset.

E. Diagnostic Criteria

On top of the standard metrics, we formalize two lightweight diagnostic criteria. Both are pure post hoc computations over the evaluation output, require no additional experiments, and their value lies in simplicity, interpretability, and auditability; they transfer to any fairness evaluation that produces group-level metrics.

False-Fairness Criterion. Difference metrics carry a degenerate blind spot: when a model makes almost no positive predictions for one subgroup, all subgroup true positive rates approach zero together, the Equal Opportunity difference approaches zero with them, and the reading resembles perfect fairness while the model fails that subgroup systematically. Let G denote the subgroups of a sensitive attribute and TPR_g the true positive rate of group g for one combination of model, attribute, and treatment. With $\Delta_{\text{EOpp}} = \max_{g \in G} \text{TPR}_g - \min_{g \in G} \text{TPR}_g$, the combination is flagged as false fairness when

$$\min_{g \in G} \text{TPR}_g < \tau \quad \text{and} \quad \Delta_{\text{EOpp}} < \tau, \quad \tau = 0.05. \quad (1)$$

The threshold is an operational choice that encodes near-zero recall for some subgroup together with a gap small enough to read as fair. The criterion targets the collapse toward all-negative prediction under low prevalence; it is insensitive to the symmetric all-positive collapse under high prevalence, a boundary treated as a finding and a limitation in Sections IV and V.

Model-Manufactured Disparity Metric. An observed selection-rate gap mixes two sources: the base-rate difference the data carry, and the share the model adds on top. Let SR_g denote the selection rate of group g and π_g its true positive-class prevalence. For each combination of model, attribute, and treatment, the metric computes

$$A = \underbrace{\left(\max_g \text{SR}_g - \min_g \text{SR}_g \right)}_{\text{observed gap}} - \underbrace{\left(\max_g \pi_g - \min_g \pi_g \right)}_{\text{true prevalence gap}}. \quad (2)$$

Amplification $A > 0$ states that the observed selection-rate gap exceeds the base-rate gap; $A < 0$ states that the output gap falls below it. The decomposition is descriptive rather than causal; the conditions under which a positive value admits no unfairness reading are stated in the limitations. The metric also records the subgroups with the largest and smallest selection rates, which makes the direction of amplification traceable to concrete populations.

IV. PERFORMANCE EVALUATION

A. Experimental Setup

All experiments run in one software environment: Python 3.11 with scikit-learn, XGBoost, Fairlearn, and AIF360, under

a fixed random seed of 42. The matrix of three classifiers, Baseline plus four treatments, and all sensitive attributes yields 45 configurations on NHANES and 30 each on Adult Income and German Credit. Every fairness gap and performance figure is computed on the 20% held-out test set, and every gap is the maximum minus the minimum across subgroups. The two diagnostic criteria produce one record per configuration. Code and result files are archived in the project repository for reproducibility.

B. RQ1: Class Imbalance, Disparity Magnitude, and Mitigation Difficulty

Baseline disparity structure contrasts sharply across the three datasets. On the low-prevalence anchor NHANES (prevalence 0.13), gaps are the largest: the Equal Opportunity difference on race reaches 0.640 for Logistic Regression and 0.677 for XGBoost, and on income it spans 0.279 to 0.471. On mid-range Adult Income (0.24), the difference on sex spans 0.133 to 0.192 and on race peaks at 0.327. On the high-prevalence anchor German Credit (0.70), the difference on sex stays between 0.055 and 0.080, the smallest among the comparable binary attributes, while the age attribute reaches 0.375. Table III lists the full baseline readings.

Mitigation cost contrasts in the same direction. On NHANES, in-processing ExponentiatedGradient cuts the Equal Opportunity difference on income by 0.333 on average, at a mean F1 cost of 0.164 and a mean recall cost of 0.267; on race its mean F1 cost is 0.097. On Adult Income, no method exceeds a mean F1 cost of 0.036 on any attribute, and on German Credit the bound is 0.035. When positives are scarce, training under an equalized odds constraint removes a large share of positive predictions outright, and the fairness gain is paid in collapsed recall (Fig. 2).

The extreme form appears in the NHANES configuration of Logistic Regression, income, and ExponentiatedGradient: on the 1,091-instance test set the model issues a single positive prediction, itself a false positive in the Above threshold subgroup, and misses all 145 actual positives, so recall drops to 0.000. The aggregate fairness metrics simultaneously report near-perfect values (Equal Opportunity difference 0.000, Demographic Parity difference 0.001). The False-Fairness Criterion flags this configuration ($\min_g \text{TPR}_g = 0.000$, $\Delta_{\text{EOpp}} = 0.000$), the only flag among all 105 configurations. The flag set is insensitive to the choice of τ : thresholds of 0.03 and 0.10 identify the same single configuration, as every other zero-recall configuration carries a gap above 0.16.

German Credit provides the reverse control. With positives as the majority class, the minimum subgroup recall never falls below 0.375 across all 30 configurations and no flag fires. A separate check of subgroup true negative rates (minimum 0.333) confirms that no all-positive collapse occurred; the criterion itself could not have detected one, because its recall floor is calibrated for the all-negative direction only. Section IV-F returns to this boundary.

The answer to RQ1 on the three anchor datasets: the low-prevalence anchor exhibits both the largest baseline disparities

TABLE III
BASELINE EQUAL OPPORTUNITY DIFFERENCE PER DATASET, ATTRIBUTE,
AND CLASSIFIER

Dataset	Attribute	LR	RF	XGB
NHANES	gender	0.161	0.166	0.131
	income	0.471	0.279	0.424
	race	0.640	0.450	0.677
Adult	sex	0.150	0.192	0.133
	race	0.327	0.086	0.125
German	sex	0.080	0.080	0.055
	age group	0.107	0.375	0.327

and the highest mitigation cost, constrained mitigation carries a risk of collapse toward all-negative prediction when positives are scarce, and the high-prevalence anchor shows the smallest binary-attribute gaps at negligible mitigation cost.

The OULAD education data set, which has an incidence of roughly 0.47 and an almost balanced split, corresponds to the medium imbalance setup. No configuration on OULAD triggers the False-Fairness Criterion, indicating no degenerate collapse occurred. The value of recall is more than 0.65 in all combinations of model and mitigation technique, with the lowest value of recall being 0.658. The biggest drop seen in the F1 score compared to the baseline is 0.021, occurring in the case of Random Forest in region below ThresholdOptimizer.

C. RQ2: Attribute Cardinality, Mitigation Effect, and Cross-Model Stability

On binary attributes, mitigation behaves cleanly. On NHANES gender, the four treatments cut the Equal Opportunity difference by 0.073 to 0.105 on average at a mean absolute F1 change of at most 0.008. On Adult sex, ExponentiatedGradient, Reweighting, and ThresholdOptimizer deliver mean reductions of 0.148, 0.128, and 0.127 at a mean F1 cost of at most 0.036 (Fig. 3).

On the five-category race attributes, the same methods change character. On NHANES race, Reweighting achieves a mean reduction of only 0.006 and Suppression widens the gap by 0.019 on average; on Adult race, ThresholdOptimizer widens the Equal Opportunity difference by 0.100 on average.

In multi-category settings the gap equals the distance between the two extreme subgroups, and neither a reweighting scheme built on a privileged-versus-rest dichotomy nor a threshold rule tuned to a global constraint controls those extremes directly, so the reduction stalls or reverses. Stability across classifiers divides along the same binary versus multi-category boundary, and on the multi-category side the divergence is wide enough to reverse conclusions. On NHANES race, ExponentiatedGradient cuts the Equal Opportunity difference of Logistic Regression from 0.640 to 0.161, a reduction of roughly three quarters; the same method moves XGBoost only from 0.677 to 0.628. ThresholdOptimizer shows the same divide: from 0.640 down to 0.210 on Logistic Regression, but 0.633 on XGBoost. The Logistic Regression results alone would certify ExponentiatedGradient as an effective treatment for the race disparity, and the XGBoost results alone would

certify the opposite. On Adult race the divide appears with reversed roles: ExponentiatedGradient pushes XGBoost to 0.046, below the 0.05 level, while Logistic Regression stalls at 0.208, so the judgment of whether the disparity is mitigated to an acceptable level flips with the base classifier. Binary attributes show no such split: after mitigation, the cross-model range of the Equal Opportunity difference stays within 0.098 on NHANES gender and within 0.041 on Adult sex, and the reading from any one classifier points to the same conclusion.

The attribute related to the OULAD region provides us with data that resembles what we have seen in case of a higher-cardinality sensitive attribute, but this time there are 13 unique regions. When applying ThresholdOptimizer, the picture that is observed with classifiers becomes completely different: Equal Opportunity difference decreases from 0.232 to 0.131 in case of Logistic Regression; for XGBoost, it drops from 0.161 to 0.133; yet, in case of Random Forest, the gap between Equal Opportunities grows drastically to 0.488 (compared to 0.140). The Reweighting algorithm produces insignificant results as well – Equal Opportunity difference in case of Logistic Regression decreases from 0.232 to 0.219, and in case of Random Forest, it drops from 0.140 to 0.121, whereas in case of XGBoost it grows slightly from 0.161 to 0.178. All in all, these results confirm the instability of the mitigation in case of higher-cardinality sensitive attributes due to the strong dependence on classifiers.

Subgroup sample size is one direct source of this instability. On Adult race, the Amer-Indian-Eskimo subgroup (96 test instances) shows a baseline true positive rate spread of 0.385 across the three classifiers, and the true positive rate of the Other subgroup (82 instances, ten actual positives) moves in steps of 0.100 between configurations; on German Credit, the young subgroup holds 28 instances with sixteen actual positives, so its true positive rate can only take values in steps of one sixteenth, and on OULAD the oldest age band holds only 39 instances.

A between-group gap built from such point estimates is highly sensitive to the flip of individual samples, and any conclusion that optimizes or evaluates against it needs the subgroup sizes attached. The answer to RQ2: as cardinality rises from binary to five categories, the disparity reduction of mitigation weakens or reverses, conclusions flip across base classifiers, and the fragility of small-subgroup point estimates adds further noise to multi-category readings.

D. RQ3: Base-Rate Gaps and Model Amplification

The Model-Manufactured Disparity Metric decomposes each observed gap into a data-inherent share and a model-added share. In the examined datasets, positive amplification dominates at baseline: on NHANES the mean over classifiers reaches 0.328 on income and 0.295 on race, and in the strongest case Logistic Regression turns the true income base-rate gap of 0.133 into an output selection-rate gap of 0.588, a model-added share of 0.456 (Fig. 4). On Adult Income the baseline means are 0.151 on sex and 0.107 on race. On German Credit the values sit near zero (0.049 on age group)

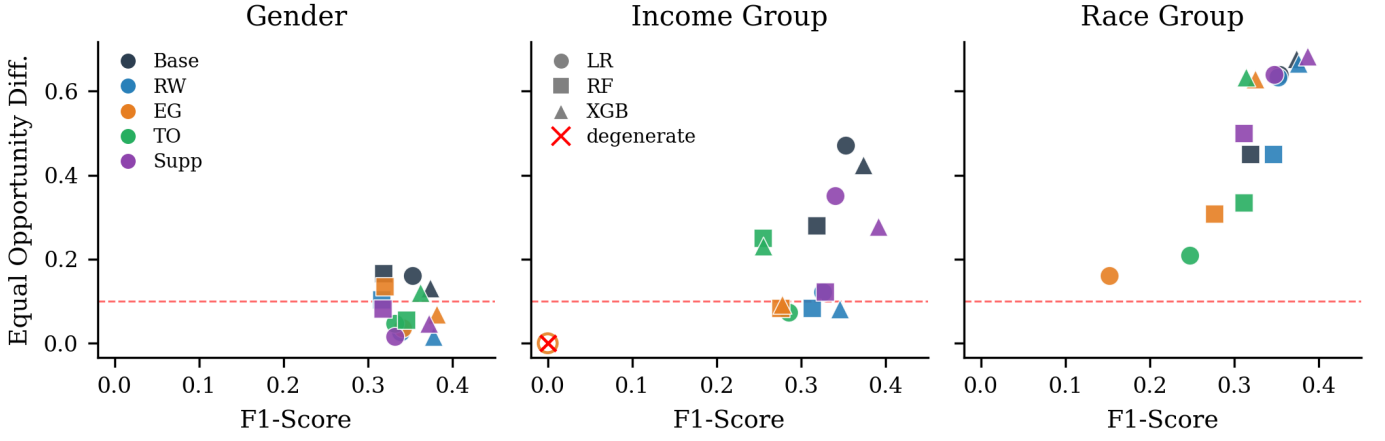


Fig. 2. Fairness-performance trade-off on NHANES: Equal Opportunity difference against F1 for every classifier and treatment, one panel per sensitive attribute, the dashed line a 0.1 reference level. Hollow markers denote degenerate configurations; the flagged Logistic Regression collapse on income appears as the hollow marker at zero F1 and zero gap.

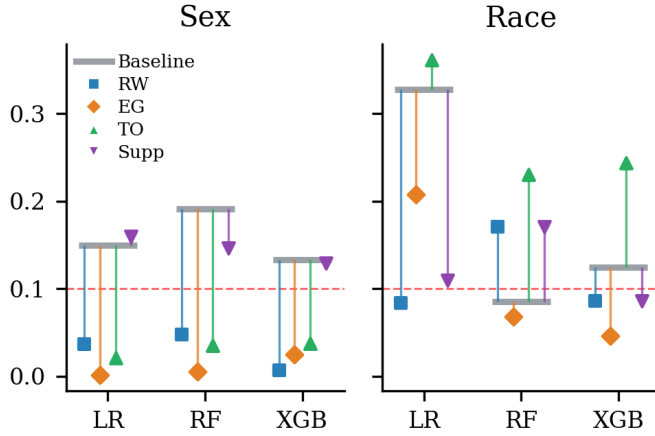


Fig. 3. Equal Opportunity difference per classifier and treatment on Adult Income. Gray bars mark the baseline level, colored markers the level after each treatment, the dashed line a 0.1 reference level. The binary sex attribute admits a consistent reduction across classifiers; on the five-category race attribute the outcome diverges by classifier and treatment.

or below it (-0.021 on sex). Read dataset by dataset, the observed gap exceeds the base-rate gap at the low- and mid-prevalence anchors, and no such excess appears at the high-prevalence anchor.

Treatments split in their effect on the model-added share. ExponentiatedGradient and ThresholdOptimizer push amplification to zero or below (mean -0.095 on NHANES income under ExponentiatedGradient, -0.089 on Adult sex under ThresholdOptimizer), so the model-added share is removable in principle, and on Adult sex the removal costs at most 0.036 in mean F1. One annotation is required: the seemingly perfect correction on NHANES income under ExponentiatedGradient contains the degenerate configuration flagged in Section IV-B, where the correction arrives through collapsed recall, and the two criteria act as mutual controls at exactly this point. Reweighting is strongly attribute-dependent, with

amplification pushed to 0.022 on NHANES income but left almost untouched at 0.281 on NHANES race. Suppression retains most of the model-added share: 0.212 on NHANES income, 0.218 on NHANES race, and 0.139 on Adult sex. Proxy features keep carrying the sensitive information after the column is removed, and the metric turns the failure of this widely assumed remedy into a measurable quantity. Negative amplification, which appears after the strongly constrained treatments, states that the output gap falls below the data-inherent one; it marks a deviation in the opposite direction rather than a neutral state.

The answer to RQ3, stated as a descriptive observation on the examined datasets: in most configurations at the low- and mid-prevalence anchors, the observed selection-rate gap exceeds the true base-rate difference; constraint-based and threshold-based mitigation remove the model-added share at moderate performance cost under mid-range prevalence, the correction can arrive through degeneracy under low prevalence, and removal of the sensitive column does not remove the model-added share.

E. Cross-Dataset Observations and Extensions

Gender is the only sensitive attribute with a comparable structure across the three datasets, and it carries the most consistent pattern in the study. Baseline Equal Opportunity differences span 0.131 to 0.166 on NHANES, 0.133 to 0.192 on Adult Income, and 0.055 to 0.080 on German Credit, and mitigation brings the difference to at or near 0.05 in the majority of classifier and method combinations on all three datasets at negligible F1 cost. One pipeline and one set of methods produce readings of consistent direction on three datasets with distinct characteristics, which supports the transferability of the evaluation and mitigation workflow and of the two criteria. Fig. 5 compares baseline and best post-mitigation gender gaps per classifier across all four datasets.

As seen in the OULAD dataset, disparity at the baseline level showed considerable differences depending on the sen-

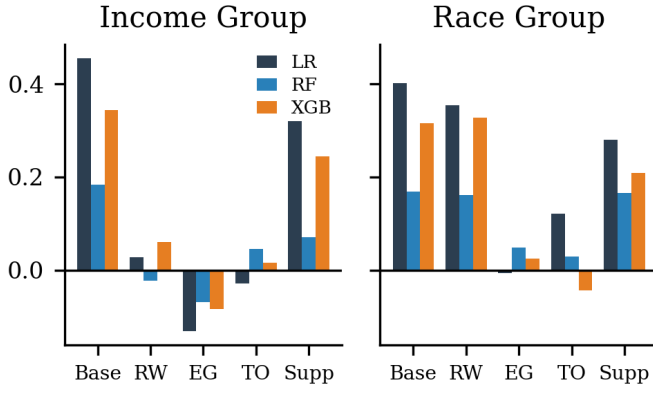


Fig. 4. Amplification across treatments on NHANES for the income and race attributes. Baseline and Suppression retain a large model-added share; ExponentiatedGradient and ThresholdOptimizer remove it.

sitive attribute being analyzed. The gender showed relatively lesser disparity than others while region and age-band had higher disparity at the baseline level, where Equalized Odds achieved a value of 0.467 for the age-band. Some of the mitigation methods were able to reduce disparities with various levels of success; for example, the use of XGBoost along with ThresholdOptimizer led to a reduction of Demographic Parity of age-band from 0.267 to 0.024 while keeping their predictive performance almost similar. The XGBoost model continued to perform well by having baseline F1-Score of 0.710 and accuracy of 0.715.

Such a phenomenon was also noticed in the other datasets as well. The Bank Marketing dataset saw a significant reduction in demographic disparity for the age group using the ThresholdOptimizer, which brought down the value of Demographic Parity to 0.0003 for Logistic Regression. In the Credit Card Clients dataset, combining the XGBoost with Exponentiated Gradient resulted in almost zero difference in the value of Demographic Parity (0.0004), although the F1-Score of 0.534 was maintained.

F. Discussion

Three mechanisms support the observations above. The collapse of ExponentiatedGradient under low prevalence has a clear optimization origin: under an equalized odds constraint with scarce positives, all-negative prediction is a trivially feasible solution, since subgroup true positive rates and false positive rates reach zero together and satisfy the constraint, and the constrained optimizer converges there along the path of least performance loss. Aggregate difference metrics cannot separate this trivial satisfaction from genuine parity, which is exactly the blind spot the False-Fairness Criterion covers. The blind spot of the criterion itself is the symmetric one: where the majority class is positive, a model can collapse toward all-positive prediction, subgroup recalls then approach one while subgroup true negative rates fall toward zero, and a criterion built on a recall floor cannot fire by construction. German Credit did not enter this mode, its minimum subgroup true

negative rate stays at 0.333, yet the criterion could not have detected the mode if it had occurred, so its zero flags on high-prevalence data carry no protective information; a symmetric criterion requires a floor on subgroup true negative rates and remains future work.

The failure of Suppression has a proxy origin: sensitive information enters the model through correlated features, so removal of the explicit column does not cut the pathway. The failure of mitigation on multi-category attributes traces instead to the gap structure described in Section IV-C, where methods built on a privileged-versus-rest dichotomy or a global constraint leave the two extreme subgroups uncontrolled.

Subgroup size adds noise on top, and the quantized true positive rates of the small subgroups on Adult Income and German Credit show that the extreme subgroups are often also the least reliably estimated ones.

V. CONCLUSION

This paper presented a model-agnostic pipeline for group fairness evaluation and mitigation that covers detection, mitigation, and reassessment, and applied it in a systematic study on three datasets characterized through measurable properties: NHANES (positive prevalence 0.13), Adult Income (0.24), and German Credit (0.70), each under three classifiers, four mitigation methods, and one control.

Our empirical investigation is extended by introducing three more datasets to our experiments: Bank Marketing (0.12), Credit Card Clients (0.22), and OULAD (0.47). This results in an increase in the number of datasets considered for the experiment from three to six while keeping the same experimental setting as before.

The experiments answer the three research questions as follows. On class imbalance, the low-prevalence anchor combines the largest baseline disparities with the highest mitigation cost, constrained mitigation can collapse toward all-negative prediction when positives are scarce, standard difference metrics read that collapse as perfect fairness, and the False-Fairness Criterion marks the single such case among 105 configurations; the high-prevalence anchor shows the smallest binary-attribute gaps at negligible cost. On attribute cardinality, the step from binary to five categories weakens or reverses the disparity reduction of mitigation, conclusions flip with the base classifier, and the fragility of small-subgroup point estimates adds further noise. On base-rate gaps, in the examined datasets the observed selection-rate gaps at the low- and mid-prevalence anchors widely exceed the true base-rate difference, constraint-based and threshold-based mitigation remove this excess at moderate cost under mid-range prevalence, and removal of the sensitive column does not.

Four limitations bound these results.

The study covers three core datasets plus three extensions; among the core datasets each application domain contributes exactly one prevalence point, so domain and prevalence covary completely in the core comparison, and the extension datasets leave this confound largely intact. Characteristic effects and domain effects therefore cannot be separated within

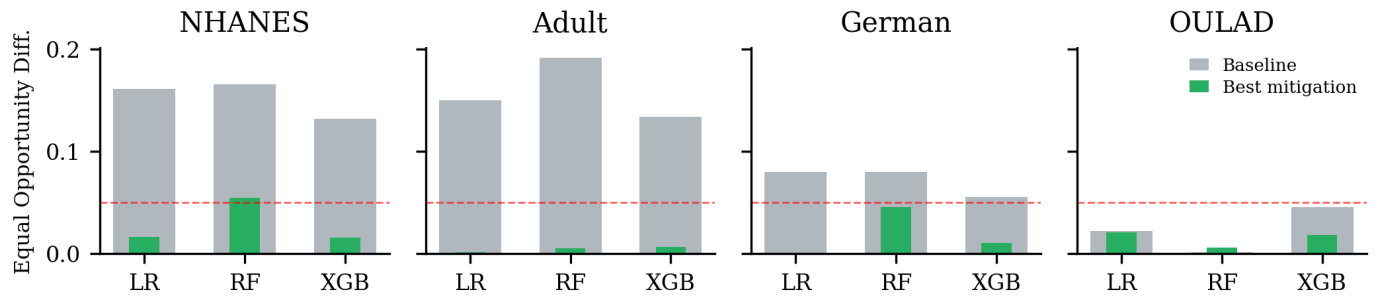


Fig. 5. Baseline and best post-mitigation Equal Opportunity difference on gender per classifier across the four datasets, the dashed line a 0.05 reference level.

this design, and every characteristic-related statement stays a descriptive observation rather than a causal claim.

Amplification is a descriptive decomposition relative to the base rate. A well-calibrated model can produce a selection-rate gap above or below the base-rate gap under different decision thresholds and error distributions, so a positive value is not by itself evidence that the model introduced unfairness; its reading depends on the training setup and the application context, and the term model-added share carries this descriptive sense throughout the paper.

The False-Fairness Criterion builds on a subgroup recall floor and is structurally insensitive to the all-positive collapse under high prevalence, so its zero flags on German Credit carry no information about that failure mode; a symmetric criterion requires a subgroup true negative rate floor.

Several subgroups are small, down to 28 instances, and their gap point estimates react strongly to the flip of individual samples. All results derive from one stratified split under seed 42, so these estimates carry additional split sensitivity, an observation that aligns with the reliability concerns of RQ2; replication across multiple splits remains open.

Future work proceeds along three lines: a symmetric criterion for the all-positive collapse with validation on high-prevalence datasets, an extension of the dataset collection that keeps the four characterization properties comparable and densifies the coverage of the coordinate system, and an integration of the two criteria into existing fairness toolchains as routine companion readings to standard difference metrics.

GENERATIVE AI DISCLOSURE

Generative AI tools were used to assist with the drafting and language refinement of this manuscript from author-provided technical content, and with debugging support and auxiliary scripts in the experimental code base. The core pipeline implementation, the research questions, the experimental design, the selection of datasets and methods, and the analysis of all results are the work of the authors. All reported numbers originate from the archived experiment outputs of the authors, and all AI-assisted content was reviewed and validated by the authors.

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